



**Cold-Formed Steel
Specialty Conference**

Elastic buckling analysis of a cold-formed steel stud column with open-source shell finite element software

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Shell finite element analysis, in the open

- Few modern shell implementations are available for public inspection, and what commercial codes do under the hood is hard to know
- A multi-year effort built a trusted, open-source shell capability in **Julia**, connected to the general-purpose finite element framework **Ferrite.jl**
- Two Mindlin shell elements are now published as packages with elastic and geometric stiffness matrices
 - **TriShellFiniteElement.jl**, a 3-node triangle
 - **QuadShellFiniteElement.jl**, a "4+2" quadrilateral
- This talk walks through one eigenbuckling analysis of a common wall stud with holes

Thanks to Sándor Ádány for the shell element formulations and MATLAB reference implementations

Two flat Mindlin shells, six dofs per node

Triangle (TriShellFiniteElement.jl)

- Constant-strain membrane; Mindlin bending with the Tessler–Hughes shear treatment (mid-side deflections condensed out)
- Shear relaxation $1/(1 + C_s\alpha)$, $C_s = 0.2$, calibrated so a simply supported plate converges to $k = 4.0$
- Hughes–Brezzi drilling stiffness, so folded plates assemble without singular matrices

Quadrilateral (QuadShellFiniteElement.jl)

- The "4+2" element of Moen and Ádány: bilinear shape functions plus two bubble functions, condensed at the element level, which relieves in-plane bending stiffness and part of the shear locking
- Same shear relaxation ($C_s = 0.1$) and drilling term
- Julia matrices reproduce the MATLAB reference to machine precision (a package test)

- Geometric stiffness from the Green–Lagrange membrane strains (N_x, N_y, N_{xy} and the gradients of all three translations); $\mathbf{K}\phi = \lambda(-\mathbf{K}_g)\phi$, and with a unit reference load the eigenvalue is the buckling load

A 362S162-33 wall stud with SFIA service holes

362S162-33

$L = 96$ in

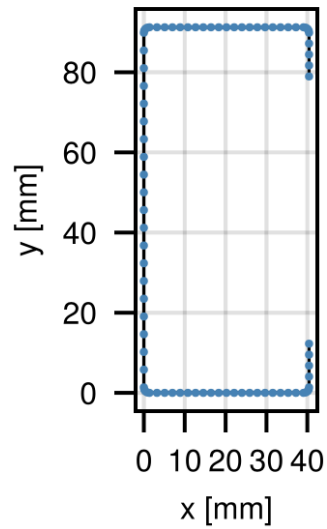
braced at midheight

pinned, warping-free ends

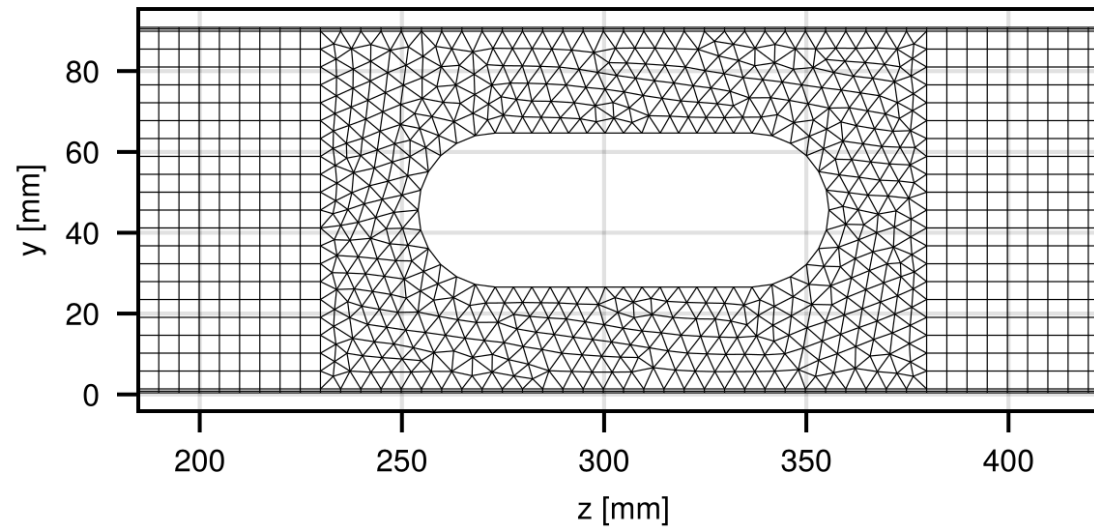
1.5×4 in holes at 24 in o.c.

- Web 3.625 in, flange 1.625 in, lip 0.5 in, design thickness **0.0346 in**; $E = 29\,500$ ksi, $\nu = 0.3$
- Brace at midheight: $L_x = L_y = L_t = 48$ in; uniform compression at both ends, $P_{\text{ref}} = 1$ N
- Four stadium-shaped holes at 12, 36, 60, and 84 in; the brace point is 12 in from the nearest holes
- Centerline polyline from CrossSectionGeometry.jl: 85 nodes, corner radius $2t$, 4 elements per lip, 20 per flange and web, 4 per corner arc
- The same polyline goes to CUFSM.jl, so the shell and finite strip analyses share their geometry

(a) cross-section, 85 nodes



(b) web face at a hole: structured quadrilaterals, triangular patch



(a) 85-node centerline polyline; (b) web mesh around a hole: structured quadrilaterals outside the 150 mm patch, triangles inside it

Degrees of freedom, boundary conditions, brace

Two element types, one DofHandler

patch-boundary nodes share dofs

```
dh = DofHandler(grid)
sq = SubDofHandler(dh, quad_cells)
ipq = Lagrange{RefQuadrilateral, 1}()
add!(sq, :u, ipq^3); add!(sq, :θ, ipq^3)
st = SubDofHandler(dh, tri_cells)
ipt = Lagrange{RefTriangle, 1}()
add!(st, :u, ipt^3); add!(st, :θ, ipt^3)
close!(dh)
```

- Ends: $u_x = u_y = 0$ at every end node, u_z and all rotations free
- Load: equal and opposite axial nodal forces by tributary centerline length, giving uniform P/A away from ends and holes
- Brace: $u_x = 0$ at the two **web-flange corners** (translation perpendicular to the web and twist), $u_y = 0$ and the axial datum $u_z = 0$ at the web mid-depth node on the symmetry axis, so no prebuckling stress is induced

Pinned warping-free ends and the midheight brace

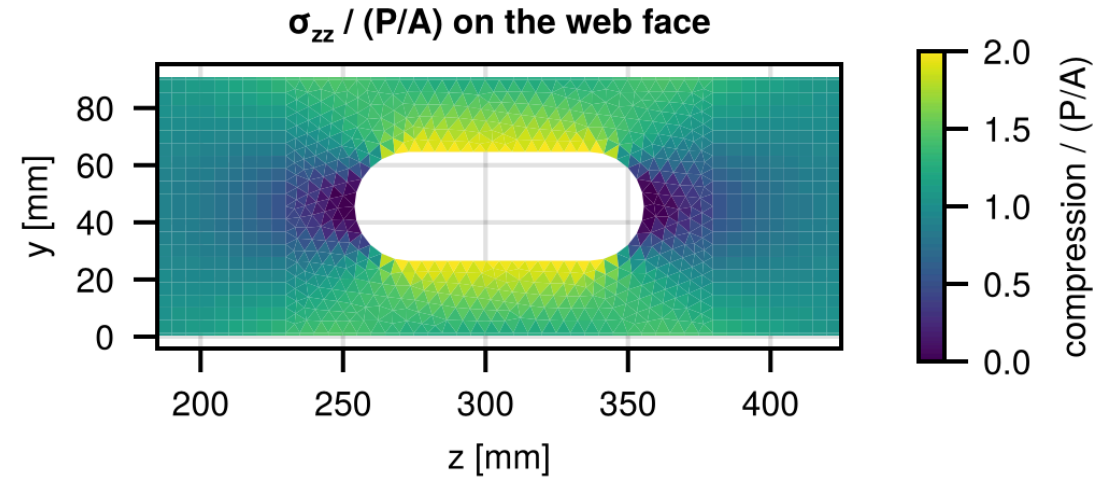
```
ch = ConstraintHandler(dh)
add!(ch, Dirichlet(:u, end_nodes,           # ux = uy = 0
  (x, t) -> [0.0, 0.0], [1, 2]))
add!(ch, Dirichlet(:u, brace_corner_nodes,   # ux = 0
  (x, t) -> [0.0], [1]))
add!(ch, Dirichlet(:u, Set([brace_web_center]), # uy = uz = 0
  (x, t) -> [0.0, 0.0], [2, 3]))
close!(ch)
```

Pre-buckling stresses and geometric stiffness

One call per package, summed

```
Kq = QuadShellFiniteElement.assemble_global_Ke!(
    allocate_matrix(dh, ch), sq, qr2, qr3,
    IP4(), IP6(), E, v, t; Cs = 0.1)
Kt = TriShellFiniteElement.assemble_global_Ke!(
    allocate_matrix(dh, ch), st, qr1, qr3,
    IP3(), IP6(), E, v, t; Cs = 0.2)
K = Kq + Kt; apply!(K, F, ch)
u = K \ F; apply!(u, ch)
σ = membrane_stresses(dh, u, E, v, t) # element frame
Kg = assemble_Kg(dh, ch, σ, t) # summed, too
```

- Static problem solved with Julia's sparse direct solver; membrane stresses recovered element by element (one per triangle, four Gauss points per quadrilateral)
- Normalized by P/A : **1.0** in the bulk, about **1.9** in the web strips beside a hole, **1.25** in the flanges at the hole plane, **0.5** just above and below the hole



Pre-buckling longitudinal membrane stress on the web face around a hole, normalized by P/A

Solving the eigenvalue problem with ARPACK

Lowest modes: $\mathbf{K}^{-1}(-\mathbf{K}_g)$, one Cholesky factorization

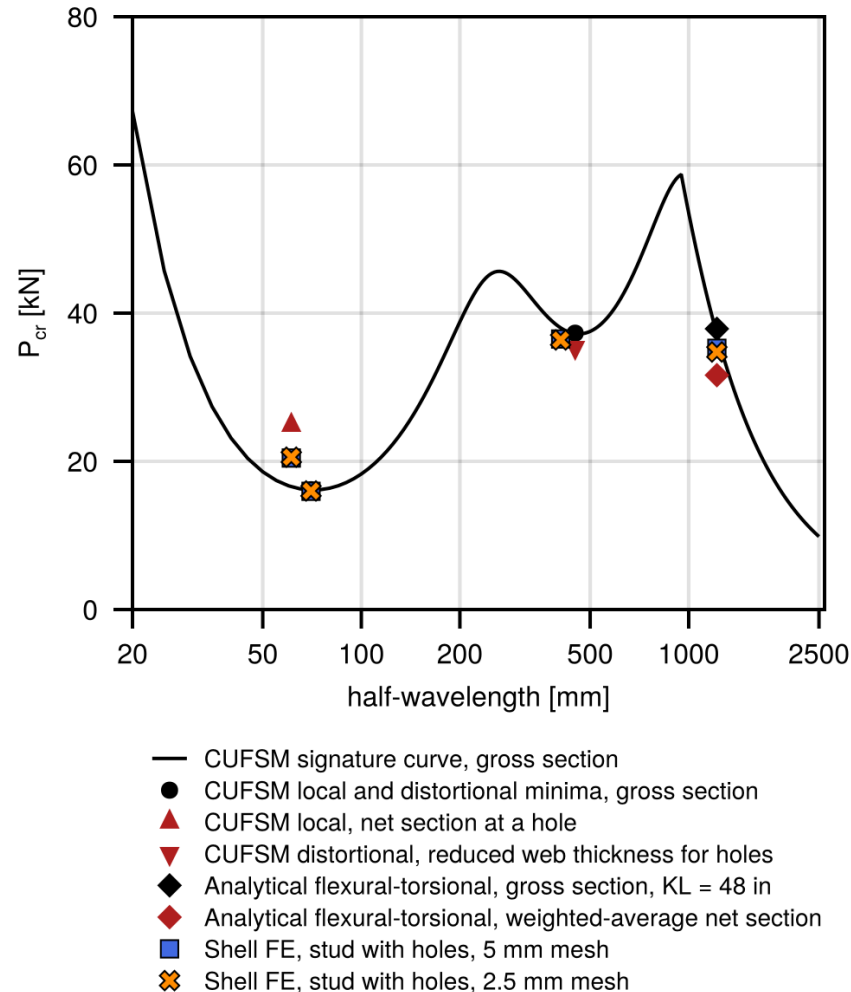
```
Kfac = cholesky(Symmetric(K)); tmp = zeros(n)
op!(y, x) = (mul!(tmp, Kg, x); y .= Kfac \ -tmp)
μ, Φ = eigs(LinearMap(op!, n; ismutating = true);
            nev = 12, which = :LM)
Pcr = 1 ./ real.(μ)      # N, because P_ref = 1 N
```

Shift-and-invert: modes nearest a target load σ

```
Mfac = ldlt(Symmetric(K + σ * Kg))      # indefinite
op!(y, x) = (mul!(tmp, Kg, x); y .= Mfac \ -tmp)
ν, V = eigs(LinearMap(op!, n; ismutating = true);
            nev = 40, which = :LM)
Pcr = σ .+ 1 ./ real.(ν)
```

- Implicitly restarted Arnoldi on $\mathbf{K}^{-1}(-\mathbf{K}_g)$: its largest eigenvalues are the reciprocals of the smallest buckling loads
- A perforated member has **hundreds of local modes** below its distortional and global loads, so the bottom of the spectrum never reaches them
- Shift-and-invert with targets from 14 to 60 kN, 40 modes per target, one indefinite factorization and about 20 s each
- Every mode is **classified automatically**: displacement ratio at the hole patches (local at a hole), rigid-body energy fraction (global: weak-axis, strong-axis, torsion), flange-lip versus web-flange corner motion (distortional), else local

Finite strip and closed-form reference values



- **CUFSM.jl, gross section:** local minimum 16.14 kN at 70 mm, distortional 37.20 kN at 460 mm
- **Hole approximations** (CeeSectionBuckling.jl, the RunToSolve SFIA evaluation procedure)
 - net section through the hole: local 24.88 kN, above the gross value, so the unperforated web governs
 - AISI S100 reduced web thickness $t_r = t(1 - L_h/L_{crd})^{1/3} = 0.807$ mm: distortional 37.30 → 35.35 kN
- **Closed form, $KL = 48$ in:** $P_{ex} = 314.6$, $P_{ey} = 56.9$, $P_t = 40.0$, flexural-torsional **37.89 kN**; weighted-average net properties 31.63 kN
- **Finite strip at one 1219 mm half-wave** (section free to distort): 37.17 kN flexural-torsional, 49.58 kN weak-axis flexure

RESULTS

Shell finite element buckling loads, kN

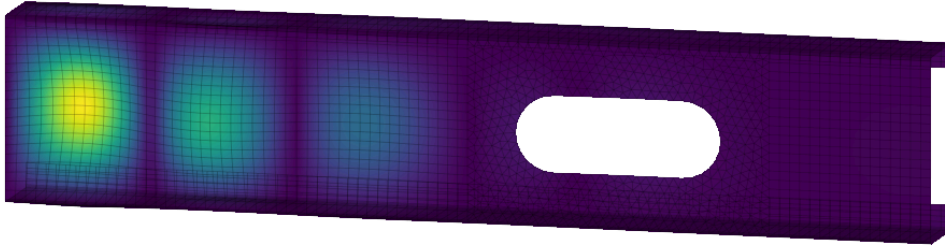
BUCKLING MODE	5 MM MESH	2.5 MM MESH	5 MM, NO HOLES	FINITE STRIP / ANALYTICAL
Local, lowest	15.95	16.00	15.79	16.14 (CUFSM gross)
Local at the holes (web strips)	20.45	20.56	–	24.88 (net section)
Distortional, 3 half-waves per segment	36.51	36.38	37.37	35.35 (reduced t), 37.30 (gross)
Distortional-hole interaction, 5 half-waves	32.63	32.33	–	–
Global, flexural-torsional	35.25	34.76	37.15	37.89 (closed form), 37.17 (CUFSM)
Global, weak-axis flexure	46.74	47.70	49.58	56.93 (Euler), 49.58 (CUFSM mode 2)

Halving the element length changes every load by less than two percent. The lowest mode is local; the governing global mode is flexural-torsional.

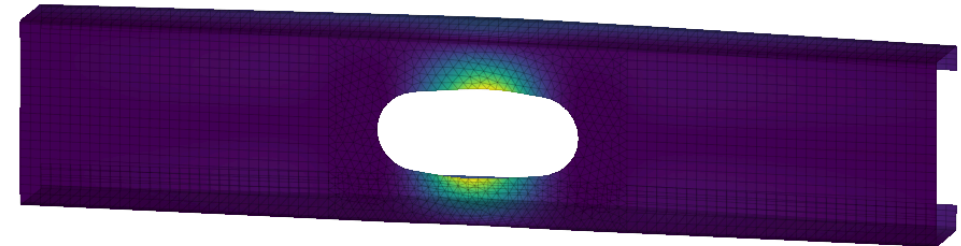
RESULTS

Local modes

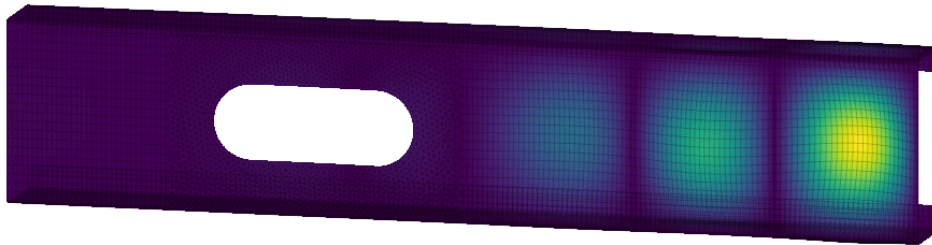
5 mm mesh, 15.95 kN (3.59 kips), $z = 0-460$ mm



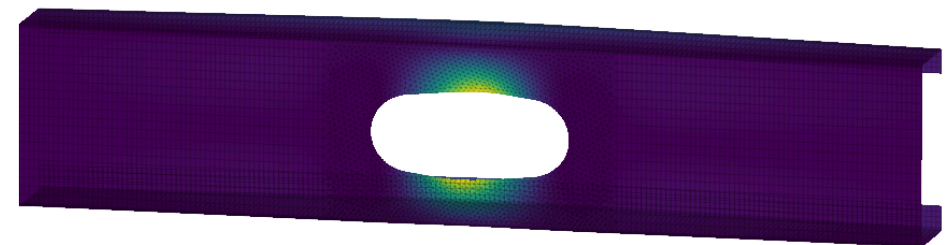
5 mm mesh, 20.45 kN (4.60 kips), $z = 1904-2364$ mm



2.5 mm mesh, 16.00 kN (3.60 kips), $z = 1978-2438$ mm



2.5 mm mesh, 20.56 kN (4.62 kips), $z = 1904-2364$ mm



Lowest local mode, 15.95 kN: the web buckles between the holes

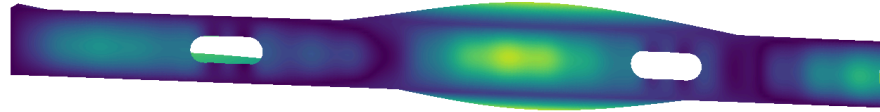
Lowest mode at a hole, 20.45 kN: the two web strips bow in opposite directions

5 mm mesh (top) and 2.5 mm mesh (bottom) in each figure; color is displacement magnitude

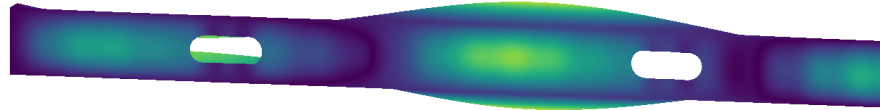
RESULTS

Distortional and global modes

5 mm mesh, 36.51 kN (8.21 kips) — 3 lip half-waves per segment



2.5 mm mesh, 36.38 kN (8.18 kips) — 3 lip half-waves per segment



Lowest distortional mode, 36.51 kN, lower 48 in braced segment: three half-waves per braced segment, the brace plane is a node

5 mm mesh, 35.25 kN (7.93 kips), full 96 in member, brace at midheight ($z = 48$ in)



Lowest global mode, 35.25 kN, full 96 in member: lateral translation plus twist, one half-wave in each braced segment, the brace at midheight is a node

Six minutes on a laptop for a quarter-million dofs

STEP	5 MM MESH	2.5 MM MESH
Degrees of freedom	247 998	509 286
Assemble elastic stiffness	6 s	9 s
Static solve	3 s	6 s
Assemble geometric stiffness	2 s	3 s
Cholesky factorization of K	10 s	20 s
12 lowest modes (ARPACK)	9 s	18 s
13 shift-and-invert windows, 40 modes each	356 s	903 s

- Apple M3 Max laptop, sparse factorizations on 7 threads, less than 2 GB of memory
- Each shift-and-invert window is one indefinite factorization and about 27 s; the sweep yields about 190 distinct modes between 16 and 65 kN
- Julia compiles the packages on first use, roughly a minute per session

Documentation and validation live on GitHub

- Each repository README documents the element mechanics, the calling sequence with Ferrite.jl, and the default parameters; docstrings cover the public functions
- The **test suites are the validation record**. For the triangle:
 - element symmetry, rigid body modes, thickness independence
 - uniform compression reproducing PL/EA
 - simply supported plate converging to $k = 4.0$, clamped plate to $k \approx 10.07$
 - buckling load invariant to the orientation of the plate in space
 - torsion of flat and folded strips and of a rounded lipped channel reproducing the thin-walled torsion constant within a few percent
- For the quadrilateral: element-by-element comparison with the MATLAB reference, plus the plate bending, membrane, and column buckling benchmarks of the earlier paper
- `Pkg.test()` on either package regenerates every number in this list



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